

Multiple Choice (10 marks)

1. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T . The average total energy of the oscillator is
 - (a) $(1/2) k T$
 - (b) kT
 - (c) $(3/2) k T$
 - (d) $3 kT$
2. The eigenvalues of a Hermitian operator are always
 - (a) real
 - (b) imaginary
 - (c) degenerate
 - (d) linear
3. The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons
 - (a) an average of 10 times, with an rms deviation of about 4
 - (b) an average of 10 times, with an rms deviation of about 3
 - (c) an average of 10 times, with an rms deviation of about 1
 - (d) an average of 10 times, with an rms deviation of about 0.1
4. The emission spectrum of the doubly ionized lithium atom Li^{++} ($Z = 3$, $A = 7$) is identical to that of a hydrogen atom in which all the wavelengths are
 - (a) decreased by a factor of 9
 - (b) decreased by a factor of 49
 - (c) decreased by a factor of 81
 - (d) increased by a factor of 9
5. The rest mass of a particle with total energy 5.0 GeV and momentum $4.9 \text{ GeV}/c$ is approximately
 - (a) $0.1 \text{ GeV}/c^2$
 - (b) $0.2 \text{ GeV}/c^2$
 - (c) $0.5 \text{ GeV}/c^2$
 - (d) $1.0 \text{ GeV}/c^2$

True / False (10 marks)

Answer True or False

6. True or False: The time required for one revolution of a satellite is independent of the mass of the planet it orbits.
7. True or False: The electric displacement current through a surface S is proportional to the rate of change of the electric flux through S.
8. True or False: The energy required to ionize helium by removing one electron from its ground state is 24.6 eV.
9. True or False: If a particle's total energy is twice its rest energy, its relativistic momentum is equal to $mc/2$.
10. True or False: Electromagnetic radiation emitted from a nucleus is most commonly in the form of gamma rays.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the Hall coefficient can be utilized to determine the sign of charge carriers in a doped semiconductor, including the underlying principles and implications of your findings.
12. Discuss the relationship between internal energy and heat transfer in an ideal gas during a constant volume process. What principles govern this interaction?

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